

ABERDEEN 35 WNW, SD, USA

WMO: 726670

Lat: 45.7114N

Lon: 99.1296W

Elev: 597

StdP: 94.36

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 54933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.4	(c) -24.3	(d) -29.6	(e) 0.3	(f) -26.3	(g) -27.3	(h) 0.3	(i) -24.1	(j) 13.3	(k) -9.3	(l) 12.1	(m) -9.9	(n) 3.3	(o) N/A	(p) 0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 31.7	(d) 21.0	(e) 29.7	(f) 20.6	(g) 28.0	(h) 19.5	(i) 23.5	(j) 28.5	(k) 22.2	(l) 27.5	(m) 21.0	(n) 26.2	(o) 4.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.8	26.3	20.5	16.3	25.3	19.2	15.0	24.0	73.1	28.4	68.1	27.6	63.7	26.2	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.9	(c) 8.6	(e) -31.3	(f) 34.8	(g) 2.9	(h) 2.2	(i) -33.4	(j) 36.4	(k) -35.1	(l) 37.7	(m) -36.8	(n) 38.9	(o) -38.9	(p) 40.5		
(a) 11.3	(b) 9.9	(c) 8.6	(e) -31.0	(f) 25.4	(g) 3.0	(h) 1.3	(i) -33.2	(j) 26.3	(k) -34.9	(l) 27.1	(m) -36.5	(n) 27.9	(o) -38.7	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-10.9	-10.4	-2.3	4.8	12.3	18.0	21.3	19.8	15.3	6.9	-1.0	-9.2
	DBStd	12.85	7.57	7.24	7.70	5.86	4.95	3.53	3.07	3.28	4.47	4.99	6.43	6.74
	HDD10.0	2868	648	572	386	172	34	1	0	0	8	121	332	594
	HDD18.3	4928	907	805	639	406	194	47	9	21	112	354	581	852
	CDD10.0	1213	0	0	5	16	105	240	349	304	166	26	2	0
	CDD18.3	232	0	0	0	0	8	37	100	67	19	1	0	0
	CDH23.3	2234	0	0	3	12	111	349	915	611	217	16	0	0
	CDH26.7	685	0	0	0	2	21	86	310	203	62	3	0	0
Wind	WSAvg	3.9	4.0	4.0	4.4	5.0	4.4	3.4	3.0	3.1	3.7	4.1	3.9	3.8
Precipitation	PrecAvg	531	13	15	24	42	78	97	78	60	52	45	16	12
	PrecMax	666	47	36	53	88	158	171	215	114	144	146	54	34
	PrecMin	341	2	2	3	11	25	30	29	15	10	4	0	1
	PrecStd	83	10	8	15	20	40	36	44	28	36	38	15	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	9.0	20.8	25.1	28.9	31.5	35.1	34.6	31.8	25.4	17.7	8.2
		MCWB	2.6	4.9	11.1	12.7	17.1	20.1	21.4	21.0	19.4	14.9	10.2	4.2
	2%	DB	3.4	4.7	16.0	20.6	26.5	29.1	31.9	31.3	28.3	20.9	13.5	4.1
		MCWB	0.9	1.8	9.0	10.8	16.7	19.8	21.5	20.9	18.9	12.3	7.5	1.1
	5%	DB	1.7	2.0	12.0	17.3	23.8	27.1	29.9	28.8	25.7	18.0	10.5	1.7
		MCWB	0.0	0.2	6.8	9.3	15.2	18.5	21.3	20.0	17.2	11.0	6.0	-0.1
	10%	DB	-0.1	0.0	8.2	14.6	21.2	25.3	28.1	26.7	23.4	15.2	7.7	-0.3
		MCWB	-1.2	-1.3	4.7	7.8	13.9	17.9	20.6	19.0	16.1	9.9	4.3	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	6.5	13.6	14.6	20.1	23.0	25.8	24.1	21.9	15.6	10.6	4.4
		MCDB	7.1	8.8	17.6	22.3	25.7	28.4	29.9	28.3	27.6	23.2	16.3	8.1
	2%	WB	1.3	2.4	10.1	11.8	18.1	21.5	24.2	22.6	20.2	13.8	8.2	1.8
		MCDB	3.3	4.8	16.7	18.3	24.8	26.7	28.7	28.7	25.2	19.2	12.8	4.3
	5%	WB	0.2	0.6	7.3	9.9	16.6	20.1	23.0	21.4	18.9	11.6	6.2	0.2
		MCDB	1.8	2.3	12.7	16.5	22.3	25.6	27.8	26.9	23.3	17.3	10.0	2.0
	10%	WB	-1.0	-1.1	5.0	8.5	14.9	19.0	21.7	20.2	17.4	10.0	4.3	-1.2
		MCDB	0.3	0.3	8.9	14.0	19.8	24.2	26.6	25.2	22.4	14.9	7.4	0.1
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.3	9.8	11.5	11.9	11.2	11.9	11.9	12.5	10.9	10.1	8.2
		MCDBR	11.1	10.6	15.9	17.2	15.3	14.1	14.6	14.9	15.5	16.3	13.9	10.7
		MCWBR	8.9	8.3	9.3	8.7	7.7	7.1	6.3	6.3	7.3	8.7	8.8	7.8
	5% WB	MCDBR	10.5	10.5	15.7	15.8	13.3	12.7	12.0	13.3	12.8	15.0	13.5	10.0
		MCWBR	8.7	8.1	9.2	8.9	7.3	7.2	6.8	6.5	7.2	8.6	8.7	7.7
Clear-Sky Solar Irradiance	taub		0.275	0.291	0.300	0.360	0.379	0.369	0.375	0.373	0.344	0.302	0.272	0.259
	taud		2.388	2.371	2.447	2.287	2.316	2.395	2.397	2.382	2.443	2.528	2.503	2.449
	Ebn at Noon		839	893	936	897	888	897	887	875	874	874	845	827
	Edh at Noon		72	89	96	124	124	116	114	111	96	76	64	61
All-Sky Solar Radiation	RadAvg		1.76	2.84	3.92	4.99	5.76	6.47	6.66	5.57	4.32	2.77	1.82	1.35
	RadStd		0.15	0.30	0.33	0.37	0.47	0.41	0.28	0.47	0.27	0.34	0.21	0.09

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page